

Electrical Circuits

Lec 4: Circuit Theorems

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Book

- **Fundamentals of Electric Circuits**
 - Charles K. Alexander
 - Matthew N. O. Sadiku
- 5th edition

Superposition

- If a circuit has two or more independent sources, one way to determine the value of a specific variable (voltage or current) is to use nodal or mesh analysis.
- Another way is to determine the **contribution of each independent source** to the variable and then add them up.
- The latter approach is known as the *superposition*.

The *superposition* principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

Steps to Apply Superposition Principle:

- 1. Turn off all **independent sources** except one source. Find the output (voltage or current) due to that active source using
- 2. Repeat step 1 for each of the other independent sources.
- 3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

Example 4.3

Use the superposition theorem to find v in the circuit of Fig. 4.6.

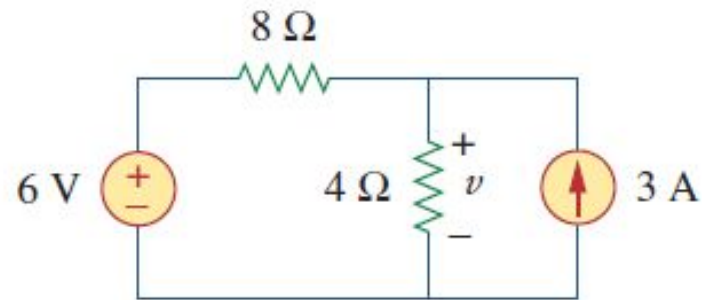
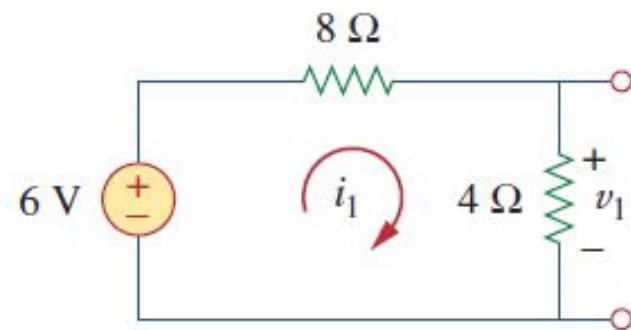


Figure 4.6
For Example 4.3.



(a)

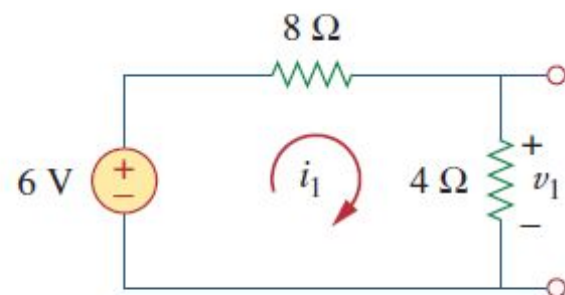
Solution:

Since there are two sources, let

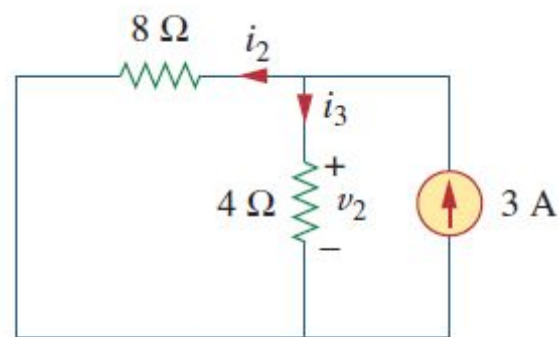
$$v = v_1 + v_2$$

where v_1 and v_2 are the contributions due to the 6-V voltage source and the 3-A current source, respectively. To obtain v_1 , we set the current source to zero, as shown in Fig. 4.7(a). Applying KVL to the loop in Fig. 4.7(a) gives

$$12i_1 - 6 = 0 \quad \Rightarrow \quad i_1 = 0.5 \text{ A}$$



(a)



(b)

Figure 4.7

For Example 4.3: (a) calculating v_1 ,
(b) calculating v_2 .

Thus,

$$v_1 = 4i_1 = 2 \text{ V}$$

We may also use voltage division to get v_1 by writing

$$v_1 = \frac{4}{4 + 8}(6) = 2 \text{ V}$$

To get v_2 , we set the voltage source to zero, as in Fig. 4.7(b). Using current division,

$$i_3 = \frac{8}{4 + 8}(3) = 2 \text{ A}$$

Hence,

$$v_2 = 4i_3 = 8 \text{ V}$$

And we find

$$v = v_1 + v_2 = 2 + 8 = 10 \text{ V}$$

Practice Problem 4.3

Using the superposition theorem, find v_o in the circuit of Fig. 4.8.

Answer: 7.4 V.

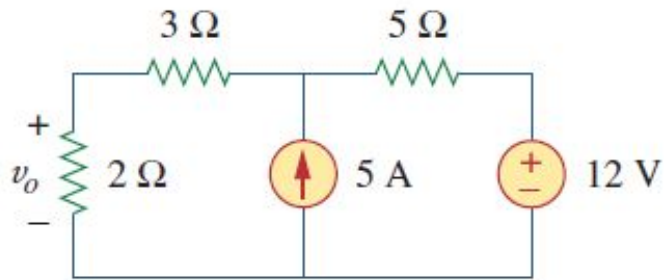


Figure 4.8

For Practice Prob. 4.3.

Example 4.4

Find i_o in the circuit of Fig. 4.9 using superposition.

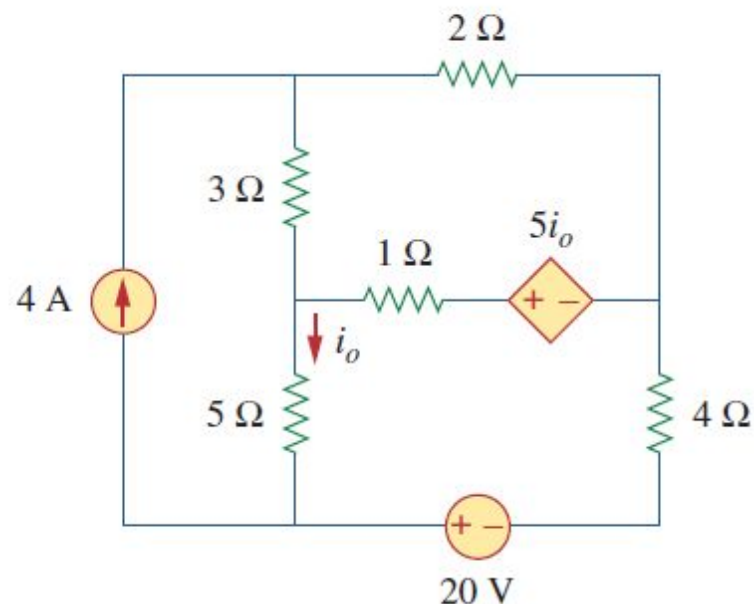


Figure 4.9
For Example 4.4.

Solution:

The circuit in Fig. 4.9 involves a dependent source, which must be left intact. We let

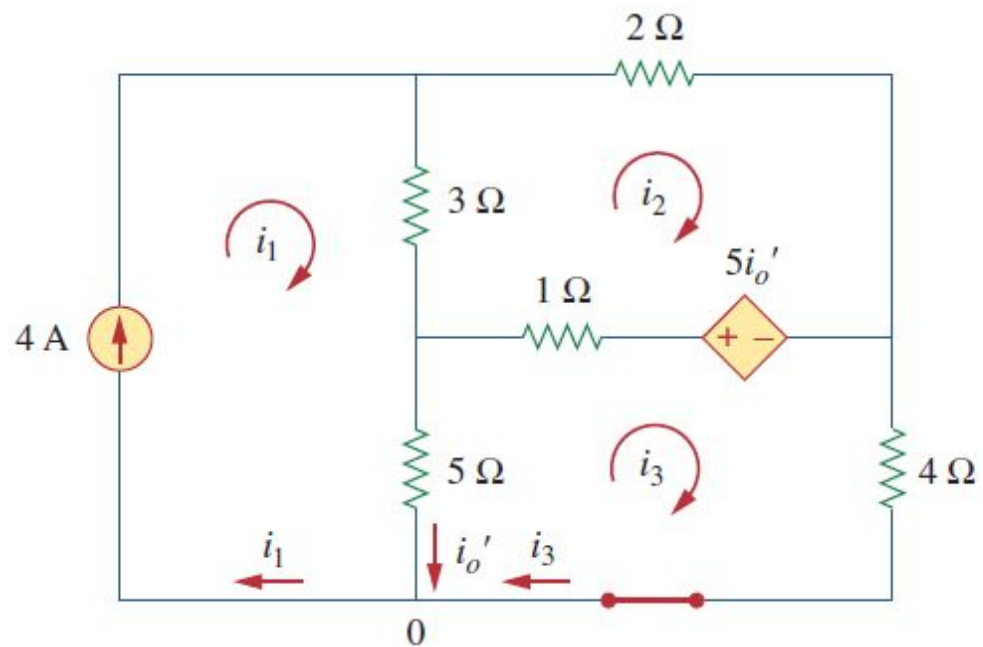
$$i_o = i'_o + i''_o \quad (4.4.1)$$

where i'_o and i''_o are due to the 4-A current source and 20-V voltage source respectively. To obtain i'_o , we turn off the 20-V source so that we have the circuit in Fig. 4.10(a). We apply mesh analysis in order to obtain i'_o . For loop 1,

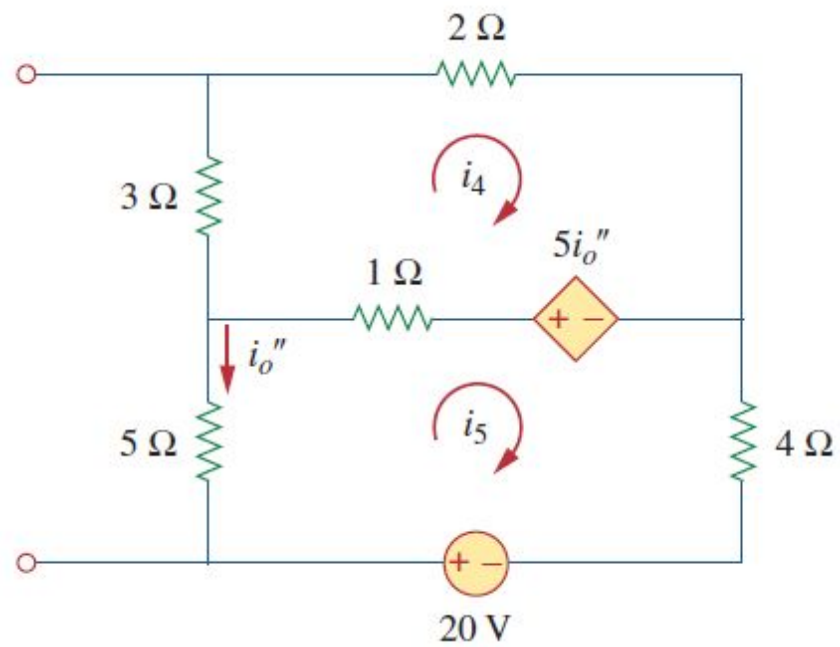
$$i_1 = 4 \text{ A} \quad (4.4.2)$$

For loop 2,

$$-3i_1 + 6i_2 - 1i_3 - 5i'_o = 0 \quad (4.4.3)$$



(a)



(b)

Figure 4.10

For Example 4.4: Applying superposition to (a) obtain i_o' , (b) obtain i_o'' .

For loop 3,

$$-5i_1 - 1i_2 + 10i_3 + 5i'_o = 0 \quad (4.4.4)$$

But at node 0,

$$i_3 = i_1 - i'_o = 4 - i'_o \quad (4.4.5)$$

Substituting Eqs. (4.4.2) and (4.4.5) into Eqs. (4.4.3) and (4.4.4) gives two simultaneous equations

$$3i_2 - 2i'_o = 8 \quad (4.4.6)$$

$$i_2 + 5i'_o = 20 \quad (4.4.7)$$

which can be solved to get

$$i'_o = \frac{52}{17} \text{ A} \quad (4.4.8)$$

To obtain i''_o , we turn off the 4-A current source so that the circuit becomes that shown in Fig. 4.10(b). For loop 4, KVL gives

$$6i_4 - i_5 - 5i''_o = 0 \quad (4.4.9)$$

and for loop 5,

$$-i_4 + 10i_5 - 20 + 5i''_o = 0 \quad (4.4.10)$$

But $i_5 = -i''_o$. Substituting this in Eqs. (4.4.9) and (4.4.10) gives

$$6i_4 - 4i''_o = 0 \quad (4.4.11)$$

$$i_4 + 5i''_o = -20 \quad (4.4.12)$$

which we solve to get

$$i''_o = -\frac{60}{17} \text{ A} \quad (4.4.13)$$

Now substituting Eqs. (4.4.8) and (4.4.13) into Eq. (4.4.1) gives

$$i_o = -\frac{8}{17} = -0.4706 \text{ A}$$

Source Transformation

A **source transformation** is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.

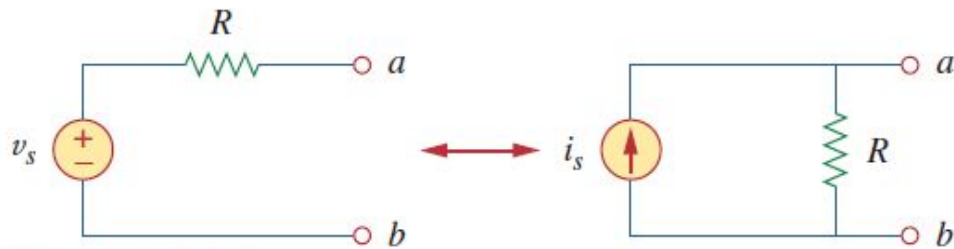


Figure 4.15

Transformation of independent sources.

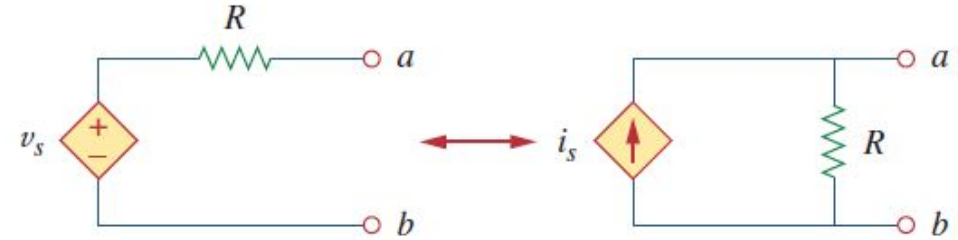


Figure 4.16

Transformation of dependent sources.

$$v_s = i_s R \quad \text{or} \quad i_s = \frac{v_s}{R}$$

Use source transformation to find v_o in the circuit of Fig. 4.17.

Example 4.6

Solution:

We first transform the current and voltage sources to obtain the circuit in Fig. 4.18(a). Combining the $4\text{-}\Omega$ and $2\text{-}\Omega$ resistors in series and transforming the 12-V voltage source gives us Fig. 4.18(b). We now combine the $3\text{-}\Omega$ and $6\text{-}\Omega$ resistors in parallel to get $2\text{-}\Omega$. We also combine the 2-A and 4-A current sources to get a 2-A source. Thus, by repeatedly applying source transformations, we obtain the circuit in Fig. 4.18(c).

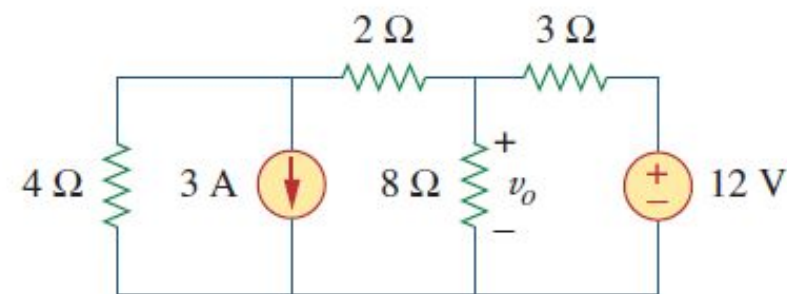
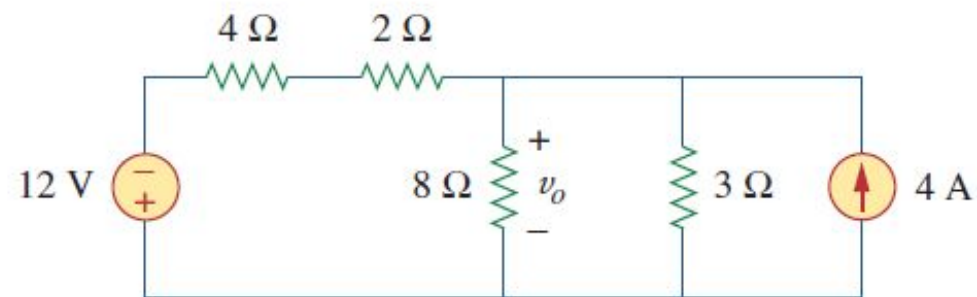
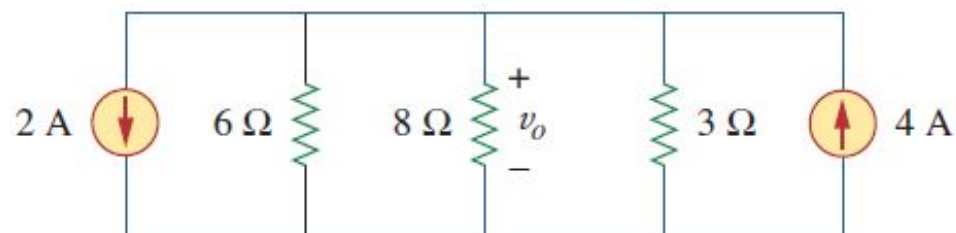


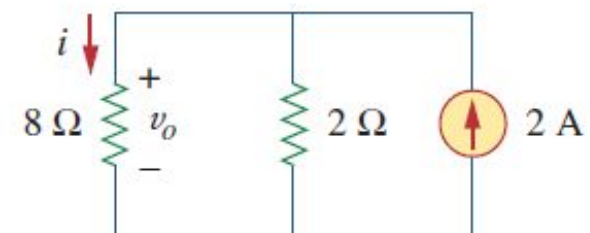
Figure 4.17
For Example 4.6.



(a)



(b)



(c)

Figure 4.18
For Example 4.6.

We use current division in Fig. 4.18(c) to get

$$i = \frac{2}{2 + 8}(2) = 0.4 \text{ A}$$

and

$$v_o = 8i = 8(0.4) = 3.2 \text{ V}$$

Alternatively, since the $8\text{-}\Omega$ and $2\text{-}\Omega$ resistors in Fig. 4.18(c) are in parallel, they have the same voltage v_o across them. Hence,

$$v_o = (8 \parallel 2)(2 \text{ A}) = \frac{8 \times 2}{10}(2) = 3.2 \text{ V}$$

Example 4.7

Find v_x in Fig. 4.20 using source transformation.

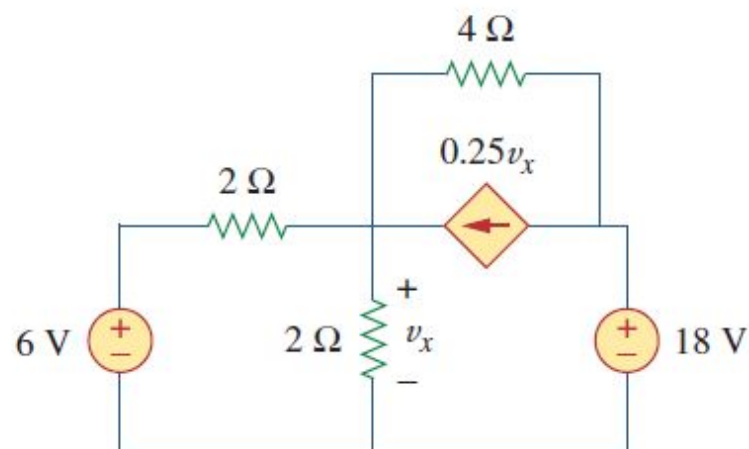


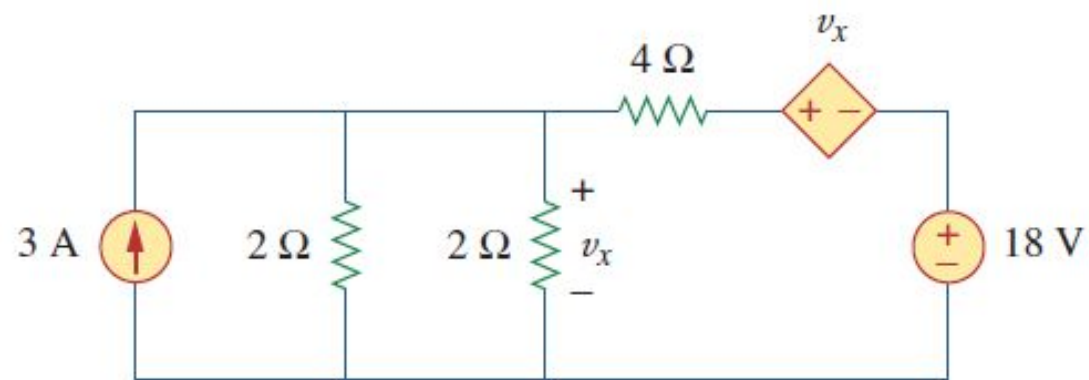
Figure 4.20

For Example 4.7.

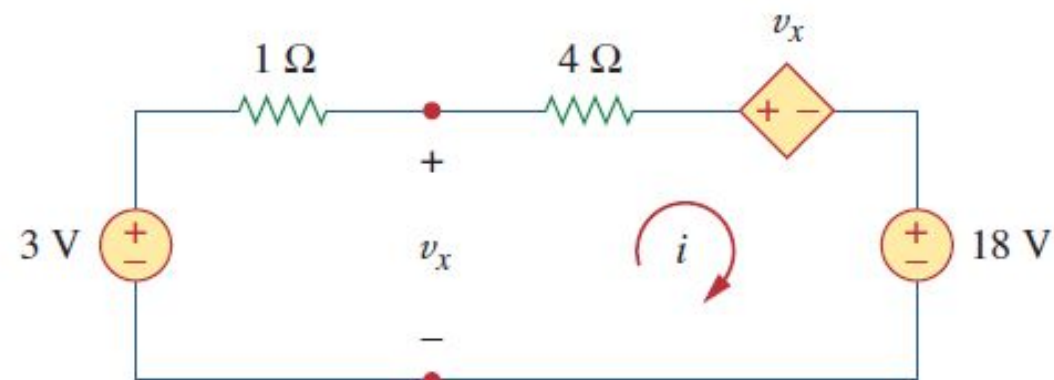
Solution:

The circuit in Fig. 4.20 involves a voltage-controlled dependent current source. We transform this dependent current source as well as the 6-V independent voltage source as shown in Fig. 4.21(a). The 18-V voltage source is not transformed because it is not connected in series with any resistor. The two 2- Ω resistors in parallel combine to give a 1- Ω resistor, which is in parallel with the 3-A current source. The current source is transformed to a voltage source as shown in Fig. 4.21(b). Notice that the terminals for v_x are intact. Applying KVL around the loop in Fig. 4.21(b) gives

$$-3 + 5i + v_x + 18 = 0 \quad (4.7.1)$$



(a)



(b)

Figure 4.21

For Example 4.7: Applying source transformation to the circuit in Fig. 4.20.

Applying KVL to the loop containing only the 3-V voltage source, the 1- Ω resistor, and v_x yields

$$-3 + 1i + v_x = 0 \quad \Rightarrow \quad v_x = 3 - i \quad (4.7.2)$$

Substituting this into Eq. (4.7.1), we obtain

$$15 + 5i + 3 - i = 0 \quad \Rightarrow \quad i = -4.5 \text{ A}$$

Alternatively, we may apply KVL to the loop containing v_x , the 4- Ω resistor, the voltage-controlled dependent voltage source, and the 18-V voltage source in Fig. 4.21(b). We obtain

$$-v_x + 4i + v_x + 18 = 0 \quad \Rightarrow \quad i = -4.5 \text{ A}$$

Thus, $v_x = 3 - i = 7.5 \text{ V}$.

Thevenin's Theorem

Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

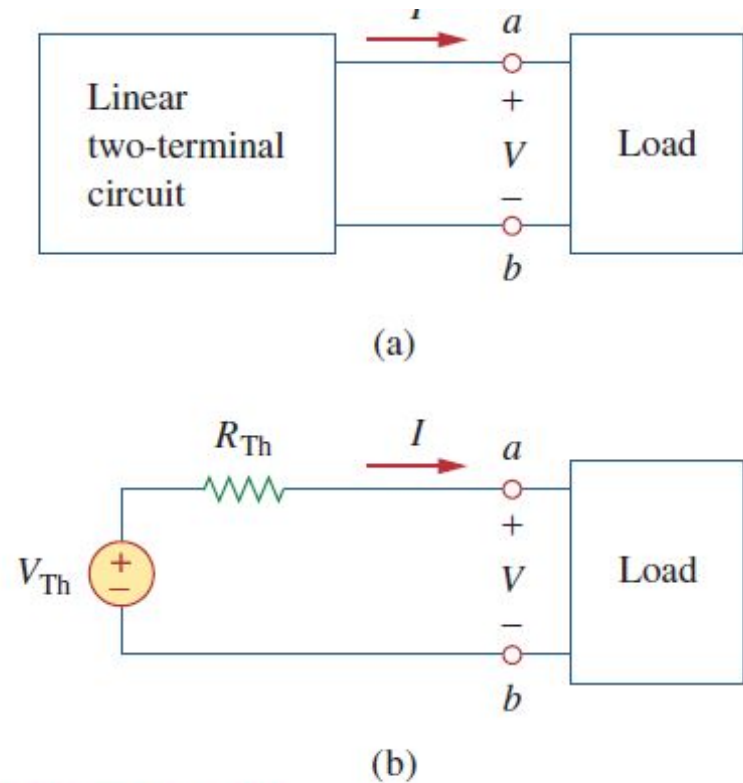


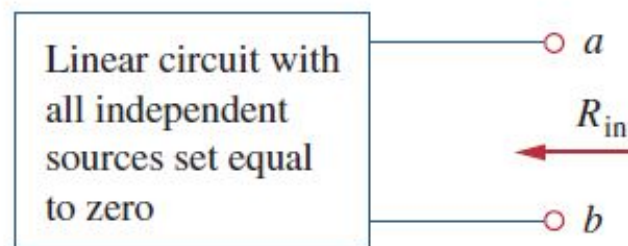
Figure 4.23

Replacing a linear two-terminal circuit by its Thevenin equivalent: (a) original circuit, (b) the Thevenin equivalent circuit.



$$V_{Th} = v_{oc}$$

(a)



$$R_{Th} = R_{in}$$

(b)

Figure 4.24

Finding V_{Th} and R_{Th} .

$$V_{Th} = v_{oc} \quad (4.6)$$

$$R_{Th} = R_{in} \quad (4.7)$$

Example 4.8

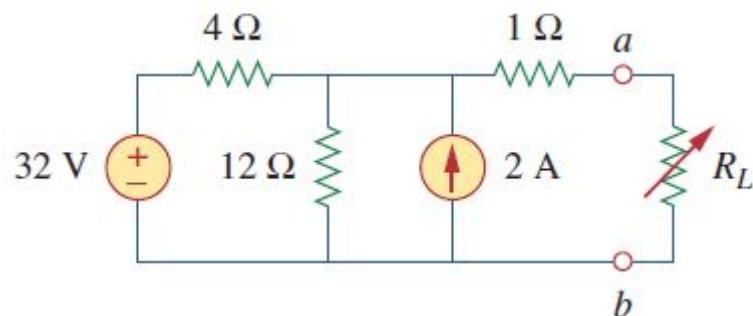


Figure 4.27
For Example 4.8.

Find the Thevenin equivalent circuit of the circuit shown in Fig. 4.27, to the left of the terminals a - b . Then find the current through $R_L = 6, 16$, and 36Ω .

Solution:

We find R_{Th} by turning off the 32-V voltage source (replacing it with a short circuit) and the 2-A current source (replacing it with an open circuit). The circuit becomes what is shown in Fig. 4.28(a). Thus,

$$R_{Th} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = 4 \Omega$$

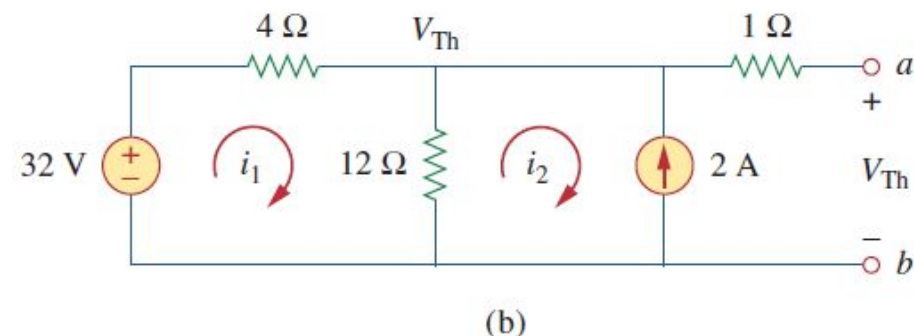
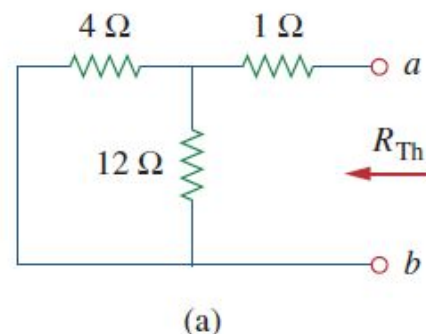


Figure 4.28

For Example 4.8: (a) finding R_{Th} , (b) finding V_{Th} .

To find V_{Th} , consider the circuit in Fig. 4.28(b). Applying mesh analysis to the two loops, we obtain

$$-32 + 4i_1 + 12(i_1 - i_2) = 0, \quad i_2 = -2 \text{ A}$$

Solving for i_1 , we get $i_1 = 0.5 \text{ A}$. Thus,

$$V_{Th} = 12(i_1 - i_2) = 12(0.5 + 2.0) = 30 \text{ V}$$

Alternatively, it is even easier to use nodal analysis. We ignore the $1\text{-}\Omega$ resistor since no current flows through it. At the top node, KCL gives

$$\frac{32 - V_{Th}}{4} + 2 = \frac{V_{Th}}{12}$$

or

$$96 - 3V_{Th} + 24 = V_{Th} \quad \Rightarrow \quad V_{Th} = 30 \text{ V}$$

as obtained before. We could also use source transformation to find V_{Th} .

The Thevenin equivalent circuit is shown in Fig. 4.29. The current through R_L is

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

When $R_L = 6$,

$$I_L = \frac{30}{10} = 3 \text{ A}$$

When $R_L = 16$,

$$I_L = \frac{30}{20} = 1.5 \text{ A}$$

When $R_L = 36$,

$$I_L = \frac{30}{40} = 0.75 \text{ A}$$

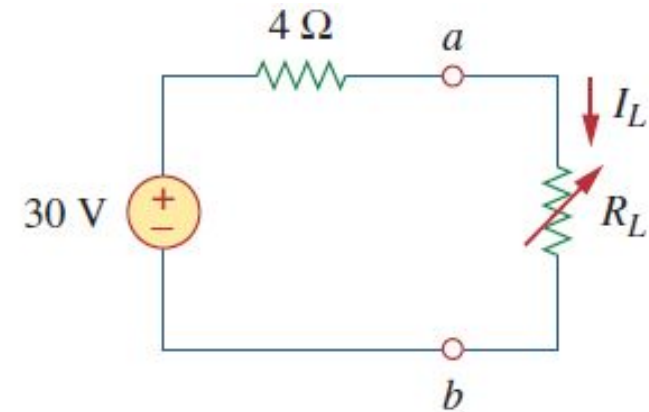


Figure 4.29

The Thevenin equivalent circuit for Example 4.8.

Practice Problem 4.8

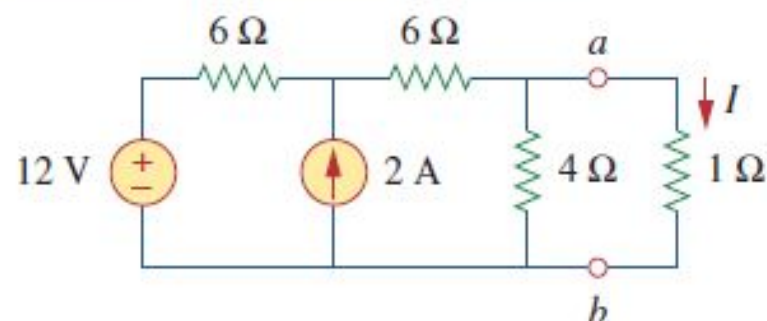


Figure 4.30

For Practice Prob. 4.8.

Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit of Fig. 4.30. Then find I .

Answer: $V_{Th} = 6 \text{ V}$, $R_{Th} = 3 \Omega$, $I = 1.5 \text{ A}$.

Example 4.9

Find the Thevenin equivalent of the circuit in Fig. 4.31 at terminals a - b .

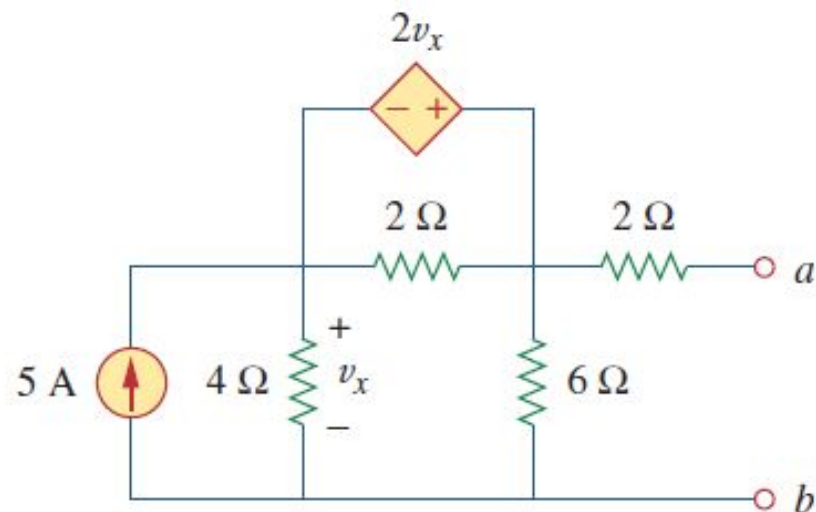
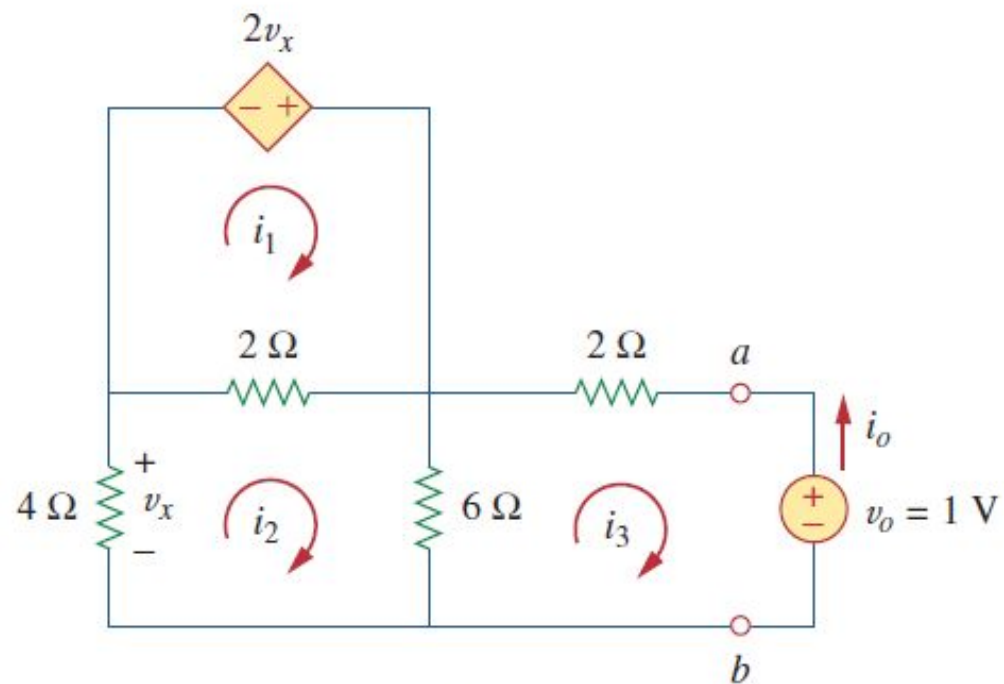


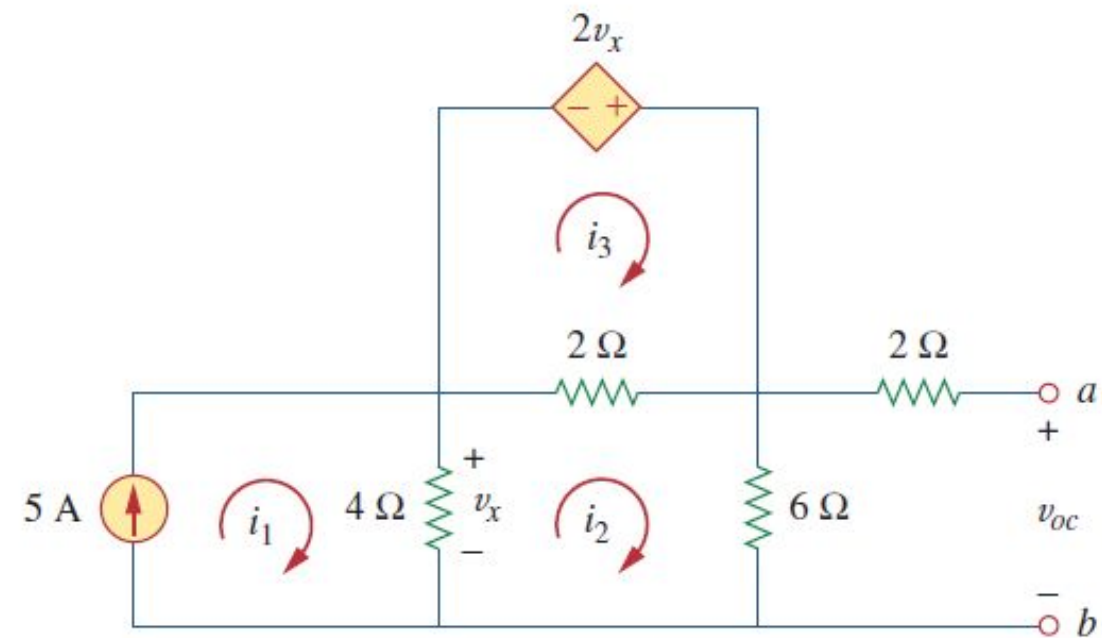
Figure 4.31
For Example 4.9.

Solution:

This circuit contains a dependent source, unlike the circuit in the previous example. To find R_{Th} , we set the independent source equal to zero but leave the dependent source alone. Because of the presence of the dependent source, however, we excite the network with a voltage source v_o connected to the terminals as indicated in Fig. 4.32(a). We may set $v_o = 1$ V to ease calculation, since the circuit is linear. Our goal is to find the current i_o through the terminals, and then obtain $R_{Th} = 1/i_o$. (Alternatively, we may insert a 1-A current source, find the corresponding voltage v_o , and obtain $R_{Th} = v_o/1$.)



(a)



(b)

Figure 4.32

Finding R_{Th} and V_{Th} for Example 4.9.

Applying mesh analysis to loop 1 in the circuit of Fig. 4.32(a) results in

$$-2v_x + 2(i_1 - i_2) = 0 \quad \text{or} \quad v_x = i_1 - i_2$$

But $-4i_2 = v_x = i_1 - i_2$; hence,

$$i_1 = -3i_2 \quad (4.9.1)$$

For loops 2 and 3, applying KVL produces

$$4i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0 \quad (4.9.2)$$

$$6(i_3 - i_2) + 2i_3 + 1 = 0 \quad (4.9.3)$$

Solving these equations gives

$$i_3 = -\frac{1}{6} \text{ A}$$

But $i_o = -i_3 = 1/6 \text{ A}$. Hence,

$$R_{\text{Th}} = \frac{1 \text{ V}}{i_o} = 6 \Omega$$

To get V_{Th} , we find v_{oc} in the circuit of Fig. 4.32(b). Applying mesh analysis, we get

$$i_1 = 5 \quad (4.9.4)$$

$$-2v_x + 2(i_3 - i_2) = 0 \quad \Rightarrow \quad v_x = i_3 - i_2 \quad (4.9.5)$$

$$4(i_2 - i_1) + 2(i_2 - i_3) + 6i_2 = 0$$

or

$$12i_2 - 4i_1 - 2i_3 = 0 \quad (4.9.6)$$

But $4(i_1 - i_2) = v_x$. Solving these equations leads to $i_2 = 10/3$. Hence,

$$V_{Th} = v_{oc} = 6i_2 = 20 \text{ V}$$

The Thevenin equivalent is as shown in Fig. 4.33.

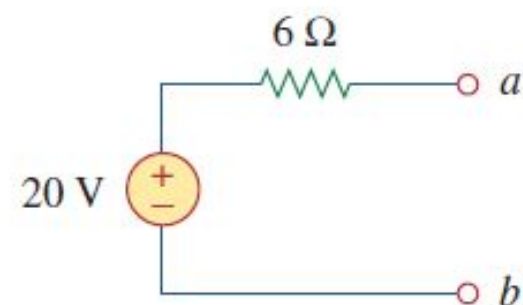


Figure 4.33

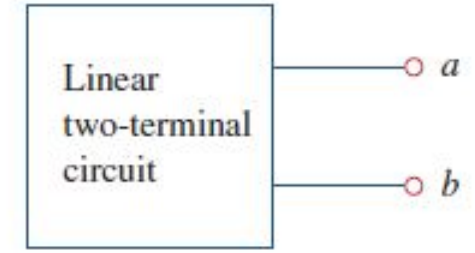
The Thevenin equivalent of the circuit in Fig. 4.31.

Norton's Theorem

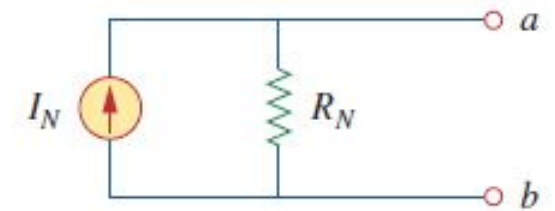
- **Norton's theorem** states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a **current source I_N in parallel with a resistor R_N** , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

$$R_N = R_{Th}$$

$$I_N = \frac{V_{Th}}{R_{Th}}$$



(a)



(b)

Figure 4.37

(a) Original circuit, (b) Norton equivalent circuit.

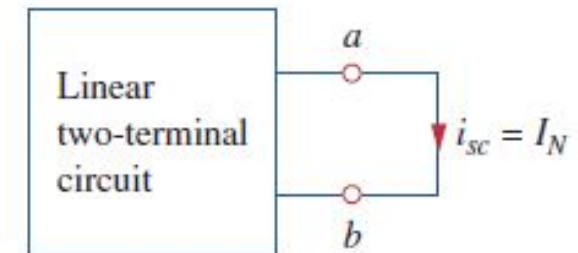


Figure 4.38

Finding Norton current I_N .

Example 4.11

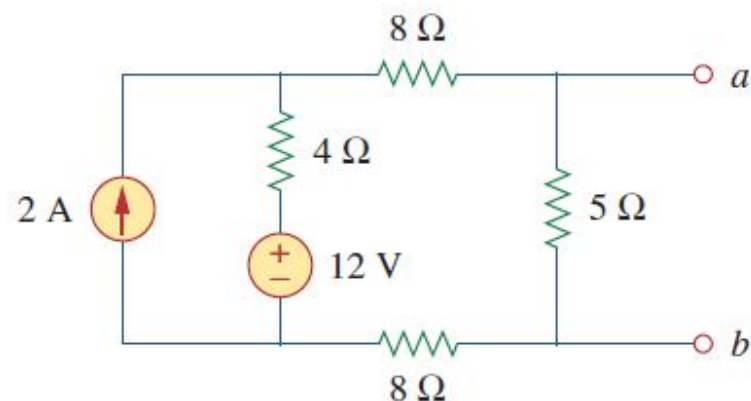


Figure 4.39

For Example 4.11.

Find the Norton equivalent circuit of the circuit in Fig. 4.39 at terminals a - b .

Solution:

We find R_N in the same way we find R_{Th} in the Thevenin equivalent circuit. Set the independent sources equal to zero. This leads to the circuit in Fig. 4.40(a), from which we find R_N . Thus,

$$R_N = 5 \parallel (8 + 4 + 8) = 5 \parallel 20 = \frac{20 \times 5}{25} = 4 \Omega$$

To find I_N , we short-circuit terminals a and b , as shown in Fig. 4.40(b). We ignore the $5\text{-}\Omega$ resistor because it has been short-circuited. Applying mesh analysis, we obtain

$$i_1 = 2 \text{ A}, \quad 20i_2 - 4i_1 - 12 = 0$$

From these equations, we obtain

$$i_2 = 1 \text{ A} = i_{sc} = I_N$$

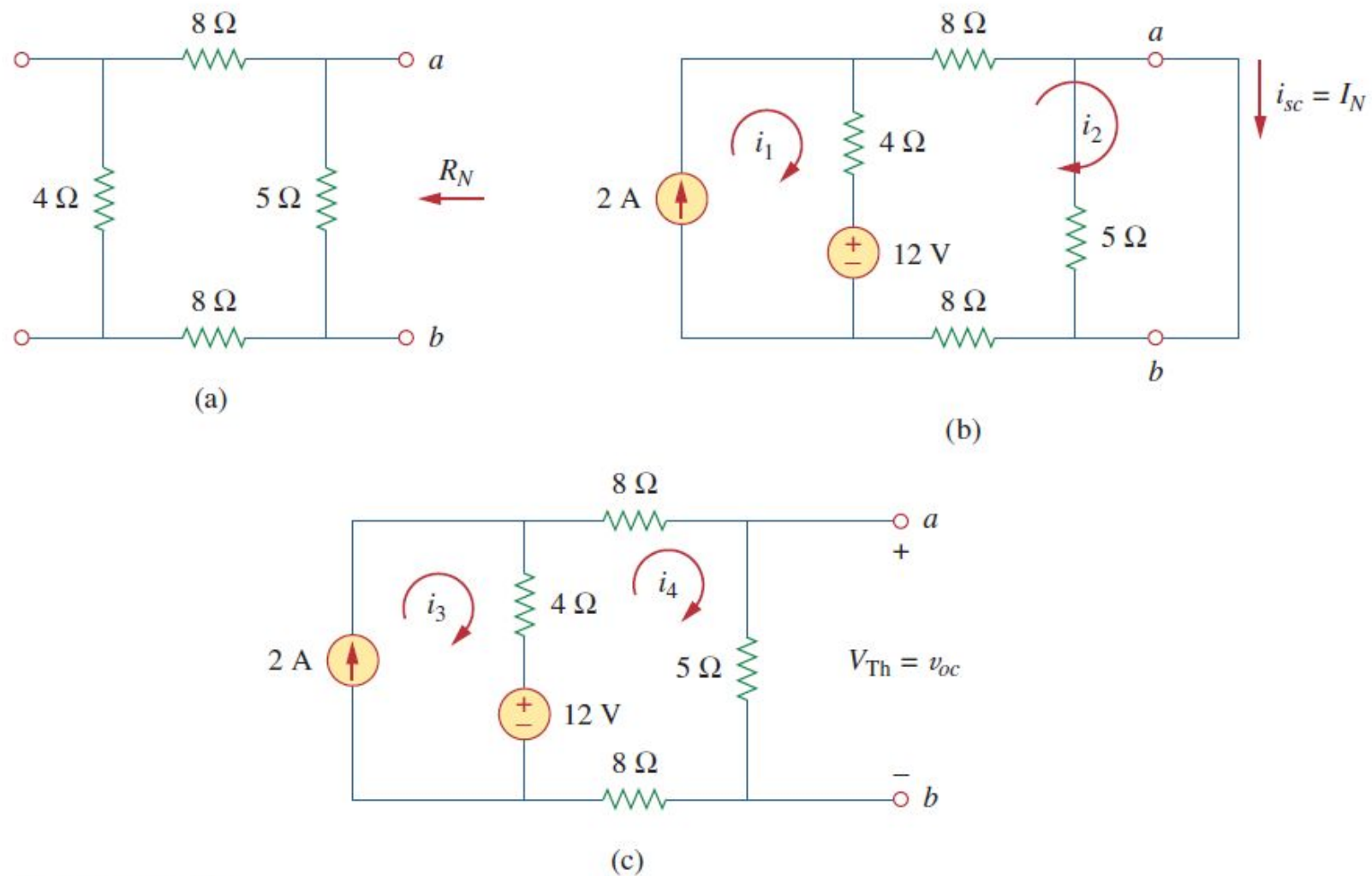


Figure 4.40

For Example 4.11; finding: (a) R_N , (b) $I_N = i_{sc}$, (c) $V_{Th} = v_{oc}$.

Alternatively, we may determine I_N from V_{Th}/R_{Th} . We obtain V_{Th} as the open-circuit voltage across terminals a and b in Fig. 4.40(c). Using mesh analysis, we obtain

$$i_3 = 2 \text{ A}$$

$$25i_4 - 4i_3 - 12 = 0 \quad \Rightarrow \quad i_4 = 0.8 \text{ A}$$

and

$$v_{oc} = V_{Th} = 5i_4 = 4 \text{ V}$$

Hence,

$$I_N = \frac{V_{Th}}{R_{Th}} = \frac{4}{4} = 1 \text{ A}$$

as obtained previously. This also serves to confirm Eq. (4.12c) that $R_{Th} = v_{oc}/i_{sc} = 4/1 = 4 \Omega$. Thus, the Norton equivalent circuit is as shown in Fig. 4.41.



Figure 4.41

Norton equivalent of the circuit in Fig. 4.39.

Maximum Power Transfer

- **Maximum power** is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \quad (4.21)$$

$$R_L = R_{Th}$$

$$p_{\max} = \frac{V_{Th}^2}{4R_{Th}}$$

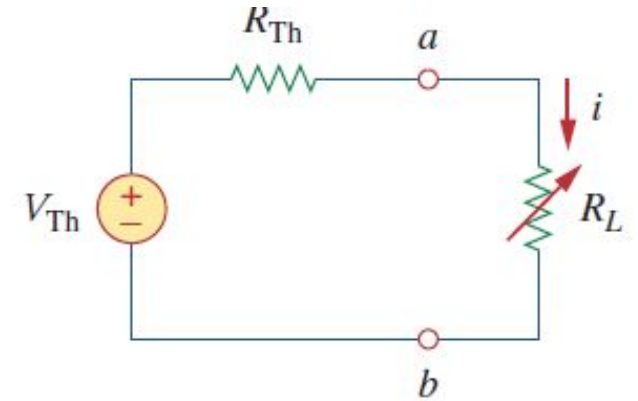


Figure 4.48

The circuit used for maximum power transfer.

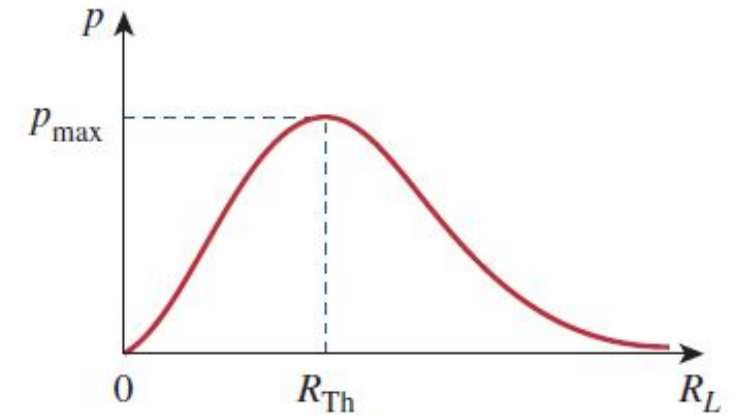


Figure 4.49

Power delivered to the load as a function of R_L .

Example 4.13

Find the value of R_L for maximum power transfer in the circuit of Fig. 4.50. Find the maximum power.

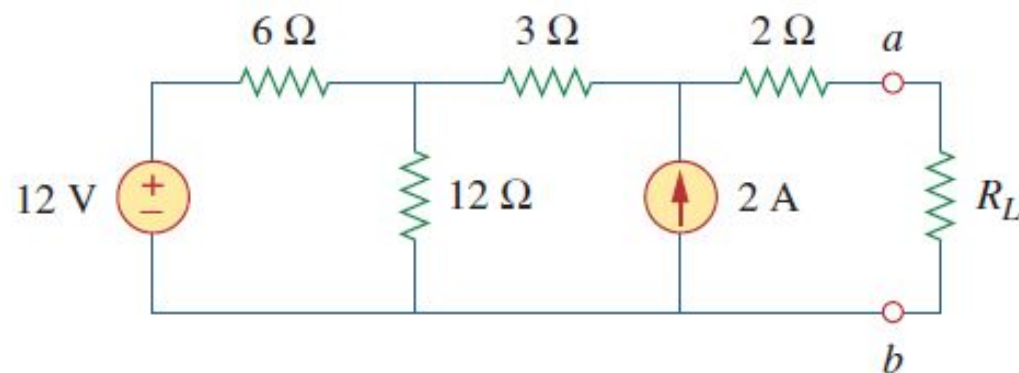
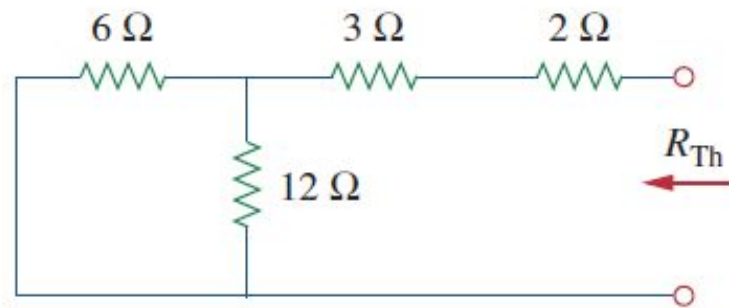


Figure 4.50
For Example 4.13.

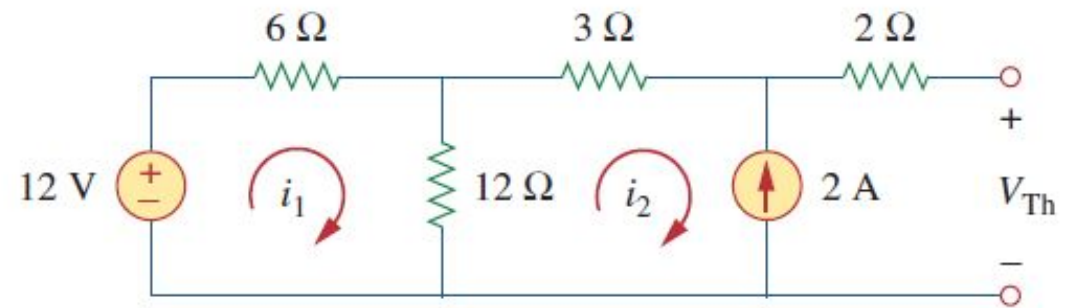
Solution:

We need to find the Thevenin resistance R_{Th} and the Thevenin voltage V_{Th} across the terminals a - b . To get R_{Th} , we use the circuit in Fig. 4.51(a) and obtain

$$R_{Th} = 2 + 3 + 6 \parallel 12 = 5 + \frac{6 \times 12}{18} = 9 \Omega$$



(a)



(b)

Figure 4.51

For Example 4.13: (a) finding R_{Th} , (b) finding V_{Th} .

To get V_{Th} , we consider the circuit in Fig. 4.51(b). Applying mesh analysis gives

$$-12 + 18i_1 - 12i_2 = 0, \quad i_2 = -2 \text{ A}$$

Solving for i_1 , we get $i_1 = -2/3$. Applying KVL around the outer loop to get V_{Th} across terminals a - b , we obtain

$$-12 + 6i_1 + 3i_2 + 2(0) + V_{\text{Th}} = 0 \quad \Rightarrow \quad V_{\text{Th}} = 22 \text{ V}$$

For maximum power transfer,

$$R_L = R_{\text{Th}} = 9 \, \Omega$$

and the maximum power is

$$p_{\text{max}} = \frac{V_{\text{Th}}^2}{4R_L} = \frac{22^2}{4 \times 9} = 13.44 \text{ W}$$

Practice Problem 4.13

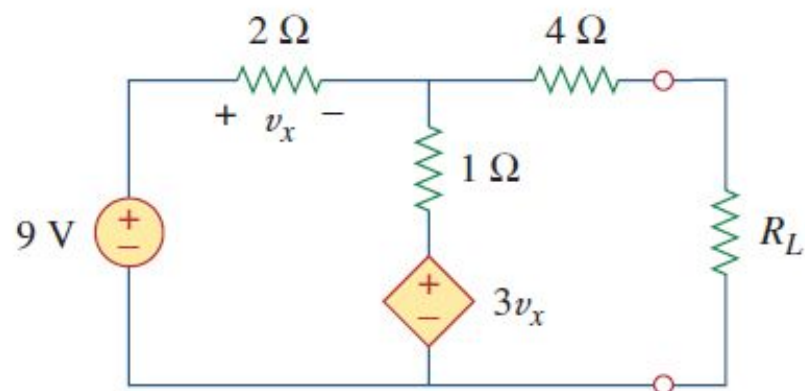


Figure 4.52

For Practice Prob. 4.13.

Determine the value of R_L that will draw the maximum power from the rest of the circuit in Fig. 4.52. Calculate the maximum power.

Answer: $4.222\ \Omega$, $2.901\ \text{W}$.